

IBVAP — Future Scope & System Enhancement Roadmap

20. Strategic Enhancements Without Breaking Current Code

1. Future Technical Enhancements (Top 10 Additions)

1. Multi-Camera RTSP Stream Ingestion

- **Description:** Upgrade single-stream file reading (**video.mp4**) to multi-threaded RTSP IP camera feed ingestion using GStreamer / OpenCV async workers.
- **Impact:** Scales coverage to dozens of simultaneous border camera channels.

2. Interactive Restricted Zone Canvas UI

- **Description:** Add a Streamlit dynamic canvas component allowing operators to draw, edit, and save polygon zones directly on top of live video feeds.
- **Impact:** Eliminates manual polygon coordinate configuration in code files.

3. Thermal / Infrared (FLIR) Model Fine-Tuning

- **Description:** Fine-tune YOLO11 weights on public thermal IR datasets (e.g. FLIR, OTCBVS).
- **Impact:** Enhances night surveillance and bad-weather target identification accuracy.

4. Permissioned Hyperledger Fabric Blockchain Integration

- **Description:** Replace single-node **blockchain_ledger.json** with a multi-node Hyperledger Fabric network running Raft consensus and chaincode smart contracts.
- **Impact:** Delivers decentralized multi-agency (e.g. BSF, Army, Intelligence) tamper-proof audit consensus.

5. IPFS (InterPlanetary File System) Evidence Storage

- **Description:** Store captured evidence JPEG snapshots on IPFS and write content multi-hashes (CIDs) to the audit log.
- **Impact:** Guarantees decentralized, unalterable visual evidence preservation.

6. Real-Time SMS / WhatsApp / Telegram Emergency Alerts

- **Description:** Integrate Twilio / Telegram Bot API to broadcast high-risk breach notifications (**HIGH**) directly to patrol officers' mobile phones within 1 second of breach.
- **Impact:** Drastically reduces tactical response latency.

7. TensorRT Model Acceleration on NVIDIA Jetson Edge Nodes

- **Description:** Export YOLO11 PyTorch model weights to NVIDIA TensorRT (**.engine** format) for edge deployment on Jetson Orin Nano hardware.
- **Impact:** Increases inference frame rates to 60+ FPS at low wattage.

8. Drone / PTZ Camera Auto-Tracking Handshake

- **Description:** Send pan-tilt-zoom (PTZ) optical movement vectors or drone flight commands toward detected breach coordinates (x, y) .
- **Impact:** Automates active target tracking beyond static camera fields of view.

9. AI Anomaly & Loitering Behavior Analysis

- **Description:** Add temporal sequence modeling (LSTM / Transformer) on target trajectories to detect suspicious loitering or crawling behaviors prior to zone entry.
- **Impact:** Provides predictive threat warnings before an intrusion occurs.

10. Role-Based Access Control (RBAC) & OAuth2 Authentication

- **Description:** Implement JWT/OAuth2 login with granular roles (**Operator**, **Commander**, **Auditor**).
- **Impact:** Enforces strict operational security governance across dashboard actions.